Literature

Wiki *Enhanced dissipation for passive scalars*

11 November 2026

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1 Reviews

Reviews (2).

- A. L. Mazzucato, Y. Feng and C. Nobili, *Enhanced dissipation by advection and applications to PDEs*, Applied Mathematical Modelling **150** (2026), Part A, 116310, arXiv:2501.17695.
- M. Coti Zelati, G. Crippa, G. Iyer and A. L. Mazzucato, *Mixing in incompressible flows: transport, dissipation, and their interplay*, Notices of the American Mathematical Society **71** (2024), 593–604, arXiv:2308.00358.

2 Deterministic enhanced dissipation

Shear, radial and other symmetric flows (19).

- S. He, *Localized Enhanced Dissipation: A Hypocoercivity Approach*, preprint (2026), arXiv:2603.14657.
- J. Benthaus, G. M. Coclite and C. Nobili, *Mixing and enhanced dissipation in a time-translating shear flow*, preprint (2026), arXiv:2603.14624.
- J. Benthaus and C. Nobili, *Enhanced dissipation via time-modulated velocity fields*, Evolution Equations and Control Theory **15** (2026), 21–50, arXiv:2501.16905.
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- J. Bedrossian, S. He, S. Iyer and F. Wang, *Pseudo-Gevrey smoothing for the passive scalar equations near Couette*, Journal of Functional Analysis **289** (2025), 110987, arXiv:2405.19233.
- D. Coble and S. He, *A note on enhanced dissipation of time-dependent shear flows*, Communications in Mathematical Sciences **22** (2024), 1685–1700, arXiv:2309.15738.
- M. Coti Zelati, M. Dolce and C. Lo, *Diffusion enhancement and Taylor dispersion for rotationally symmetric flows in discs and pipes*, Journal of Mathematical Fluid Mechanics **26** (2024), Article 12, arXiv:2305.18162.
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- M. Coti Zelati and T. Gallay, *Enhanced dissipation and Taylor dispersion in higher-dimensional parallel shear flows*, Journal of the London Mathematical Society **108** (2023), 1358–1392, arXiv:2108.11192.
- D. Albritton, R. Beekie and M. Novack, *Enhanced dissipation and Hörmander's hypoellipticity*, Journal of Functional Analysis **283** (2022), 109522, arXiv:2105.12308.
- S. He, *Enhanced dissipation, hypoellipticity for passive scalar equations with fractional dissipation*, Journal of Functional Analysis **282** (2022), 109319, arXiv:2103.07906.
- M. Coti Zelati and T. D. Drivas, *A stochastic approach to enhanced diffusion*, Annali della Scuola Normale Superiore di Pisa, Classe di Scienze **22** (2021), 811–834, arXiv:1911.09995.
- M. Coti Zelati, *Stable mixing estimates in the infinite Péclet number limit*, Journal of Functional Analysis **279** (2020), 108562, arXiv:1909.01310.
- M. Coti Zelati and M. Dolce, *Separation of time-scales in drift-diffusion equations on R^2*, Journal de Mathématiques Pures et Appliquées **142** (2020), 58–75, arXiv:1907.04012.
- M. Beck, O. Chaudhary and C. E. Wayne, *Rigorous justification of Taylor dispersion via center manifolds and hypocoercivity*, Archive for Rational Mechanics and Analysis **235** (2020), 1105–1149, arXiv:1804.06916.
- J. Bedrossian and M. Coti Zelati, *Enhanced dissipation, hypoellipticity, and anomalous small noise inviscid limits in shear flows*, Archive for Rational Mechanics and Analysis **224** (2017), 1161–1204, arXiv:1510.08098.
- M. Beck, O. Chaudhary and C. E. Wayne, *Analysis of enhanced diffusion in Taylor dispersion via a model problem*, Hamiltonian Partial Differential Equations and Applications, Fields Institute Communications **75** (2015), pp. 31–71, arXiv:1501.05877.
- A. Kolmogoroff, *Zufällige Bewegungen (Zur Theorie der Brownschen Bewegung)*, Annals of Mathematics (2) **35** (1934), 116–117, doi:10.2307/1968123.
- W. Thomson (Lord Kelvin), *Stability of fluid motion: rectilineal motion of viscous fluid between two parallel planes*, Philosophical Magazine (5) **24** (1887), 188–196, doi:10.1080/14786448708628078.

Other smooth flows (4).

- M. Dolce, C. J. P. Johansson and M. Sorella, *Dissipation enhancing properties for a class of Hamiltonian flows with closed streamlines*, Communications in Partial Differential Equations **50** (2025), 301–352, arXiv:2407.06884.
- T. M. Elgindi, K. Liss and J. C. Mattingly, *Optimal enhanced dissipation and mixing for a time-periodic, Lipschitz velocity field on T2*, Duke Mathematical Journal **174** (2025), 1209–1260, arXiv:2304.05374.
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- G. Iyer and H. Zhou, *Quantifying the dissipation enhancement of cellular flows*, SIAM Journal on Mathematical Analysis **55** (2023), 6496–6516, arXiv:2209.11645.

Rough velocity fields (5).

- M. Romito and L. Roveri, *Instantaneous Total Enhanced Dissipation For Very Rough Shear Flows*, preprint (2025), arXiv:2510.27331.
- T. M. Elgindi and K. Liss, *Norm growth, non-uniqueness, and anomalous dissipation in passive scalars*, Archive for Rational Mechanics and Analysis **248** (2024), arXiv:2309.08576.
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General criteria and limits (7).

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- P. Constantin, A. Kiselev, L. Ryzhik and A. Zlatoš, *Diffusion and mixing in fluid flow*, Annals of Mathematics **168** (2008), 643–674, arXiv:math/0509663.

3 Stochastic enhanced dissipation

Random velocity fields (2).

- W. Cooperman, G. Iyer and S. Son, *A Harris theorem for enhanced dissipation, and an example of Pierrehumbert*, Nonlinearity **38** (2025), 045027, arXiv:2403.19858.
- J. Bedrossian, A. Blumenthal and S. Punshon-Smith, *Almost-sure enhanced dissipation and uniform-in-diffusivity exponential mixing for advection-diffusion by stochastic Navier-Stokes*, Probability Theory and Related Fields **179** (2021), 777–834, arXiv:1911.01561.

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- F. Flandoli and D. Luo, *Enhanced dissipation and Lyapunov exponents for stochastic transport-diffusion equations with small molecular diffusivity and small noise intensity*, Stochastics and Partial Differential Equations: Analysis and Computations **14** (2026), 1062–1076, doi:10.1007/s40072-026-00426-y.
- M. Coti Zelati, M. Hairer and D. Villringer, *A Stochastic RAGE Theorem and Enhanced Dissipation for Transport Noise*, preprint (2025), arXiv:2507.11422.

4 Deterministic mixing

Deterministic mixing (4).

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5 Stochastic mixing

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